

Ambulance diversions

CHL5222 Final Project (Group B)

Study description

Emergency department (ED) overcrowding is a problem in the state of California. ED overcrowding can be extremely damaging to patient health. “Ambulance diversion”, which is when hospital closes its doors and forces ambulances to bring patients to alternative hospitals is one response to ED overcrowding. The California Office of Statewide Health Planning and Development collected data on how often hospitals enacted “diversion status”, enabling researchers to investigate factors associated with increasing amounts of ambulance diversions. The data set contains the following variables from California hospitals with ED over a 3-year period (2013-2015).

Data dictionary

Table 1: Data dictionary for ambulance study

variable_name	variable_description
admit	Total number of admitted EMS visits over the year
admitrate	Rate of admission (admit/totalvisits)
county	County
diverthours	Number of hours of diversion status over the year
emslevel	EMS level (Standby, Basic, Comprehensive)
id	Hospital identifier
stations	Number of emergency department stations available for patients
totalvisits	Total number of EMS visits over the year
trauma	Trauma center designation (0 = None, Level I, II, III)
year	Year (2013-2015)

Research question

What are the factors that are associated with the number of ambulance diversion hours among California hospitals during 2013-2015?